

ATTACK PATH DISCOVERY AND SEVERITY ASSESSMENT FRAMEWORK

A PROJECT REPORT

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IN

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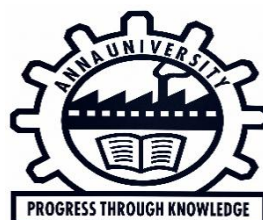
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ABSTRACT

Modern enterprise networks consist of highly interconnected systems, where a single vulnerability can be exploited to execute multi-stage cyberattacks, resulting in significant organizational impact. Traditional security tools focus on identifying individual vulnerabilities but fail to provide insights into how attackers can chain these weaknesses into realistic attack paths. To address this limitation, this paper proposes an Attack Path Discovery and Severity Assessment Framework that performs offensive security analysis through a web-based MERN application. The framework integrates and correlates outputs from widely used tools such as Nmap, OpenVAS, SearchSploit, LinPEAS, BloodHound, and CrackMapExec to provide a comprehensive view of potential attack vectors. It generates graph-based attack paths that illustrate privilege escalation and lateral movement, along with severity ranking to aid in effective risk prioritization. The proposed approach enhances security analysis by offering actionable insights without performing automated exploitation, enabling organizations to proactively strengthen their defense mechanisms against advanced threats.

திட்டச் சுருக்கம்

நவீன நிறுவன நெட்வொர்க்குகள் மிகவும் பரஸ்பர இணைக்கப்பட்ட அமைப்புகளைக் கொண்டுள்ளன. இதில் ஒரு தனிப்பட்ட பலவீனம் கூட பல கட்டங்களைக் கொண்ட சைபர் தாக்குதல்களுக்கு வழிவகுத்து, நிறுவனத்திற்கு மிகுந்த பாதிப்பை ஏற்படுத்த முடியும். தற்போதைய பாதுகாப்பு கருவிகள் ஒவ்வொரு பலவீனத்தையும் தனித்தனியாக கண்டறிந்தாலும், தாக்குதலாளர்கள் அவற்றை எவ்வாறு இணைத்து உண்மையான தாக்குதல் பாதைகளை உருவாக்குகிறார்கள் என்பதை விளக்குவதில் குறைபாடு உள்ளது. இந்த சிக்கலை தீர்க்க, இந்த ஆய்வில் MERN அடிப்படையிலான வலை பயன்பாட்டின் மூலம் செயல்படும் **Attack Path Discovery and Severity Assessment Framework** முன்மொழியப்படுகிறது. இந்த அமைப்பு Nmap, OpenVAS, SearchSploit, LinPEAS, BloodHound மற்றும் CrackMapExec போன்ற கருவிகளின் முடிவுகளை ஒருங்கிணைத்து பகுப்பாய்வு செய்கிறது. இது privilege escalation மற்றும் lateral movement ஆகியவற்றைக் காட்டும் graph அடிப்படையிலான attack paths மற்றும் severity ranking வழங்குகிறது. தானியங்கி தாக்குதலை மேற்கொள்ளாமல், முன்னெச்சரிக்கை பாதுகாப்பு நடவடிக்கைகளுக்கான பயனுள்ள பார்வையை இந்த அணுகுமுறை வழங்குகிறது.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATIONS	FULL FORM
1	AD	Active Directory
2	CVE	Common Vulnerability Exposure
3	CVSS	Common Vulnerability Severity Score
4	JSON	JavaScript Object Notation
5	JWT	JSON Web Token
6	OpenVAS	Open Vulnerability Assessment System
7	MERN	MongoDB,Express js,React js,Node js
8	LinPEAS	Linux Privilege Escalation System
9	Nmap	Network Mapping
10	IP	Internet Protocol
11	HTTP	Hyper Text Transfer Protocol
12	GUI	Graphic User Interface
13	BFS	Breadth First Search

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In today's rapidly evolving digital landscape, cyberattacks have become increasingly sophisticated, targeting complex enterprise environments such as Active Directory and interconnected network systems. Traditional security solutions primarily focus on detecting individual vulnerabilities, but they often fail to identify how multiple weaknesses can be combined by attackers to form complete attack paths. This limitation makes it difficult for organizations to fully understand their security posture and effectively prioritize defense strategies.

The Attack Path Discovery and Severity Assessment Framework is designed to address this challenge by providing a comprehensive and intelligent approach to cybersecurity analysis. The system integrates automated vulnerability scanning with advanced graph-based attack path discovery, enabling the identification of potential exploitation paths within a network.

In this framework, the enterprise environment is modeled as a graph, where nodes represent entities such as users, systems, and permissions, and edges represent possible attack transitions between them. Using this model, the system can analyze relationships and automatically discover multiple attack paths that an adversary might exploit to move laterally and reach critical assets.

Additionally, the framework performs severity assessment by evaluating factors such as vulnerability exposure, privilege escalation opportunities, and proximity to critical resources. This helps security teams prioritize high-risk attack paths rather than focusing on isolated vulnerabilities.

By combining automation, centralized data management, and visual attack path analysis, this project enhances the ability of security professionals to detect, analyze, and mitigate threats proactively, ultimately strengthening the overall cybersecurity posture of an organization.

1.2 THE MAIN FEATURES

1.2.1 Secure Authentication & Authorization

1. User registration and login system
2. Password encryption using bcrypt
3. JWT-based authentication for secure API access
4. Role-based access control

1.2.2 Target Management

1. Upload .txt files containing domains/IPs
2. Supports bulk scanning of multiple targets
3. Input validation for secure file handling

1.2.3 Automated Vulnerability Scanning

1. Integration with Nuclei scanner
2. Background scan execution (non-blocking)
3. Configurable scan parameters (rate, timeout, templates)

1.2.4 Scan Tracking & History

Real-time scan status:

1. Pending → Running → Completed/Failed
2. Maintains full scan history with timestamps
3. Enables monitoring and auditing

1.2.5 Vulnerability Analysis & Classification

1. Categorizes vulnerabilities into
2. Critical, High, Medium, Low, Informational
3. Provides detailed information:
4. Template ID, description, affected host, references

1.2.6 Severity-Based Filtering & Insights

1. Fetch vulnerabilities based on severity

2. Helps prioritize high-risk issues
3. Provides summary and breakdown of results

1.2.7 Attack Path Discovery (Graph-Based)

1. Models the network as a graph structure
2. Nodes → Users, systems, permissions
3. Edges → Attack transitions
4. Identifies possible attack paths attackers may exploit
5. Supports analysis of multi-step attacks

1.2.8 Graph Visualization of Attack Paths

1. Visual representation of attack chains
2. Helps security teams understand complex relationships
3. Highlights critical paths leading to sensitive assets

1.2.9 Severity Assessment of Attack Paths

1. Evaluates risk based on:
2. Privilege escalation
3. Vulnerability exposure
4. Distance to critical resources
5. Helps in prioritizing remediation

1.2.10 Centralized Data Storage

1. Uses LowDB (JSON database)
2. Stores users, scans, and vulnerabilities
3. Ensures persistent and structured data

1.2.11 Logging & Monitoring

1. Uses Winston logger
2. Maintains error logs and system logs
3. Supports debugging and audit tracking

1.2.12 Security Features

1. Input validation and sanitization
2. File size/type restrictions
3. Secure error handling
4. CORS protection

1.3 PURPOSE OF THE SYSTEM

The purpose of the Attack Path Discovery and Severity Assessment Framework is to provide an advanced cybersecurity solution that not only identifies individual vulnerabilities but also discovers complete attack paths using graph-based analysis.

Traditional security tools focus on isolated vulnerabilities, making it difficult for security teams to understand how attackers can combine multiple weaknesses to compromise critical systems. This project addresses that gap by modeling the enterprise environment as a graph structure, where nodes represent entities such as users, systems, and permissions, and edges represent possible attack transitions.

By visualizing these relationships, the system enables security analysts to:

1. Understand how attackers can move laterally across a network
2. Identify high-risk attack paths leading to critical assets
3. Gain better visibility into complex network security

The ultimate purpose is to enhance proactive defense strategies by combining vulnerability scanning, attack path discovery, and severity assessment into a unified system.

1.4 OBJECTIVE OF THE PROJECT

1.4.1 Discover Attack Paths Using Graph-Based Analysis

1. Model the network as a graph (nodes & edges)
2. Identify multiple possible attack paths attackers can exploit
3. Visualize relationships between users, systems, and permissions

1.4.2 Provide Graph Visualization of Security Risks

1. Represent attack paths visually for easy understanding
2. Help security teams quickly identify critical paths
3. Improve decision-making through intuitive visualization

1.4.3 Automate Vulnerability Detection

1. Use Nuclei scanner to detect vulnerabilities
2. Continuously scan multiple targets
3. Reduce manual security assessment effort

1.4.4 Perform Severity Assessment of Attack Paths

1. Evaluate risks based on:
2. Privilege escalation opportunities
3. Vulnerability exposure
4. Distance to critical assets
5. Assign severity levels to prioritize threats

1.4.5 Enable Lateral Movement Analysis

1. Identify how attackers can move across the network
2. Detect weak links between systems
3. Prevent multi-step attacks

1.4.6 Centralize Security Data

1. Store vulnerabilities, scans, and attack paths in one system
2. Provide structured and persistent data access

1.4.7 Improve Risk Prioritization

1. Highlight high-risk attack paths instead of isolated vulnerabilities
2. Help teams focus on the most dangerous threats first

1.4.8 Support Proactive Cyber Defense

1. Identify risks before exploitation

2. Strengthen critical assets and access points
3. Enhance overall organizational security posture

1.5 SCOPE OF THE PROJECT

1.5.1 Enterprise Network Security Analysis

1. Can be used to analyze organizational networks
2. Helps identify vulnerabilities and attack paths in real environments

1.5.2 Penetration Testing & Red Teaming

1. Assists ethical hackers in:
2. Identifying weak points
3. Understanding attack chains

1.5.3 Vulnerability Management Systems

1. Acts as a centralized vulnerability management platform
2. Helps organizations track and manage security issues

1.5.4 Attack Path Analysis & Risk Assessment

1. How attackers move across systems
2. Which paths are most dangerous
3. Enables risk-based prioritization

1.5.5 Security Operations (SOC) Support

1. Helps SOC teams monitor vulnerabilities
2. Improves incident response planning
3. Enhances threat analysis capabilities

1.5.6 Academic & Research Applications

1. Useful for cybersecurity research in:
2. Graph-based security analysis
3. Can be extended for academic projects and studies

1.5.7 Future Enhancements Scope

1. Threat intelligence feeds
2. CVSS scoring systems
3. AI-based attack prediction

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

In the current cybersecurity landscape, most vulnerability assessment and security analysis systems rely on traditional scanning tools such as Nmap, Nessus, and OpenVAS. These tools are primarily designed to detect individual vulnerabilities in systems, networks, and applications. While they are effective in identifying security weaknesses, they often operate in isolation and provide results in the form of static reports.

These existing systems focus mainly on listing vulnerabilities based on severity levels such as critical, high, medium, and low. However, they do not provide a comprehensive understanding of how multiple vulnerabilities can be interconnected or exploited together by attackers. As a result, security analysts are required to manually interpret the scan results and identify potential attack paths, which is both time-consuming and prone to human error.

Furthermore, traditional systems lack advanced visualization capabilities. Most tools generate text-based or tabular reports, making it difficult for users to understand complex relationships between users, systems, permissions, and vulnerabilities within a network. There is no structured mechanism to represent the enterprise environment as a connected model that reflects real-world attack scenarios.

Another limitation is the absence of centralized data management and historical tracking. Many tools do not maintain a unified database of past scans, making it difficult to analyze trends, monitor improvements, or track recurring vulnerabilities over time. Additionally, existing systems often require manual configuration and execution, limiting automation and scalability.

Overall, the existing systems are fragmented and lack the ability to provide attack path discovery, graph-based visualization, and intelligent risk prioritization, which are essential for modern cybersecurity defense.

2.1.1 Problems Of The Existing System

1. Lack of attack path discovery:

Existing tools do not identify how multiple vulnerabilities can be combined into a complete attack chain.

2. Isolated vulnerability analysis:

Vulnerabilities are detected individually without correlation or relationship analysis.

3. Manual analysis required:

Security analysts must manually interpret scan results, increasing effort and chances of error.

4. Limited visualization:

Most systems provide only text-based reports without graphical representation of attack paths.

5. Poor risk prioritization:

Focus is only on severity scores, without considering real-world exploitability or attack chains.

6. Fragmented tools:

Multiple tools are required for scanning, analysis, and reporting, leading to inefficiency.

7. Lack of centralized storage:

No proper database to store and manage scan results and history.

8. Limited automation:

Scans often require manual initiation and monitoring.

2.1.2 Limitations Of The Existing System

1. Inability to model network relationships:

Existing systems do not represent users, systems, and permissions as interconnected entities.

2. No graph-based analysis:

Lack of graph models to visualize and analyze attack paths.

3. Difficulty in identifying lateral movement:

Cannot track how attackers move across systems.

4. Scalability issues:

Handling large-scale networks and vulnerabilities becomes inefficient.

5. Lack of real-time insights:

Systems do not provide dynamic updates or continuous monitoring.

6. Dependency on manual expertise:

Requires skilled analysts to interpret results effectively.

7. No integrated solution:

Scanning, analysis, and visualization are not combined into a single framework.

2.2 PROPOSED SYSTEM

The proposed system, “Attack Path Discovery and Severity Assessment Framework,” is designed to overcome the limitations of traditional vulnerability management systems by providing an integrated, automated, and intelligent approach to cybersecurity analysis.

The system begins by allowing users to upload target files containing domains or IP addresses, which are then processed using the Nuclei vulnerability scanner. The scanning process is executed in the background, ensuring non-blocking and efficient performance. The results generated by Nuclei are stored in a structured format within a centralized database, enabling easy access and management of vulnerability data.

Unlike existing systems, the proposed framework goes beyond individual vulnerability detection by introducing attack path discovery using graph-based modeling. In this approach, the enterprise environment is represented as a graph where nodes correspond to entities such as users, systems, and services, and edges represent possible attack transitions.

By analyzing these relationships, the system can identify multiple attack paths that an attacker may exploit to move laterally across the network and reach critical assets.

The framework also incorporates graph visualization techniques to provide a clear and intuitive representation of attack paths. This visualization helps security analysts understand complex relationships within the network and quickly identify high-risk paths that require immediate attention.

In addition, the system performs severity assessment by evaluating factors such as vulnerability exposure, privilege escalation opportunities, and proximity to critical resources. This allows the system to prioritize risks more effectively compared to traditional severity-based approaches.

The proposed system includes secure authentication using JWT, ensuring that only authorized users can access the platform. It also maintains a complete scan history, enabling users to track previous scans and analyze trends over time. Logging mechanisms are implemented for monitoring system activities and debugging purposes.

Overall, the proposed system provides a unified platform that integrates vulnerability scanning, attack path discovery, severity analysis, and visualization, thereby enhancing the ability of organizations to detect, analyze, and mitigate security threats proactively.

2.2.1 Advantages Of The Proposed Systems

1. Provides attack path discovery using graph-based analysis, enabling identification of multi-step attack chains.
2. Offers visual representation of attack paths, making it easier to understand complex network relationships.

3. Automates vulnerability scanning using Nuclei, reducing manual effort and time consumption.
4. Enables centralized storage of vulnerabilities, scans, and results for efficient data management.
5. Performs severity assessment based on multiple factors, improving risk prioritization.
6. Supports real-time scan tracking and status monitoring.
7. Provides secure access using JWT-based authentication and role-based authorization.
8. Reduces dependency on multiple tools by integrating scanning, analysis, and visualization into a single system.
9. Helps security analysts identify high-risk attack paths quickly and take proactive measures.
10. Improves decision-making for remediation by focusing on critical vulnerabilities and attack chains.
11. Maintains complete scan history, enabling trend analysis and audit tracking.
12. Scalable and adaptable for different environments such as enterprise networks, cloud systems, and web applications.

CHAPTER 3

LITERATURE SURVEY

3.1 AUTOMATED VULNERABILITY ASSESSMENT USING NETWORKING TOOLS

Authors: Various cybersecurity researchers

Published: 2022

Overview:

This study focuses on automating vulnerability assessment using tools such as Nmap and OpenVAS. The system performs network scanning to identify open ports, running services, and potential vulnerabilities. The collected data is analyzed to detect security weaknesses in target systems.

3.1.1 Problem :

1. Manual vulnerability assessment is time-consuming and requires expert knowledge
2. Difficulty in identifying all possible vulnerabilities in complex systems
3. Lack of integration between scanning tools
4. Limited visibility into attack paths

3.1.2 Solution:

1. Automate scanning using tools like Nmap and OpenVAS
2. Extract service and version information
3. Map detected vulnerabilities with known CVEs
4. Provide structured vulnerability reports

3.1.3 Advantages:

1. Reduces manual effort

2. Improves detection speed
3. Provides detailed vulnerability insights
4. Enhances security assessment accuracy

3.2 ATTACK GRAPH GENERATION FOR NETWORK SCANNING ANALYSIS

Authors: Security research community

Published: 2023

Overview:

This paper introduces attack graphs as a method to visualize possible attack paths in a network. It models how an attacker can move through a system by exploiting multiple vulnerabilities.

3.2.1 Problem:

1. Security tools provide isolated results without correlation
2. Difficult to understand multi-step attacks
3. Lack of visualization for attack progression

3.2.2 Solution:

1. Generate attack graphs using collected vulnerability data
2. Represent nodes as system states and edges as attack steps
3. Correlate vulnerabilities to form attack chains
4. Use graph databases like Neo4j for visualization

3.2.3 Advantages:

1. Helps understand attacker behavior
2. Provides clear visualization of attack paths
3. Supports risk prioritization

4. Improves decision-making in security analysis

3.3 EXPLOIT INTELLIGENCE USING SEARCHSPLOIT AND CVE MAPPING

Authors: Cybersecurity researchers

Published: 2021

Overview:

This research focuses on linking identified vulnerabilities with available exploits using tools like Searchsploit. It helps security analysts understand whether a vulnerability is exploitable in real-world scenarios.

3.3.1 Problem:

1. Difficulty in identifying exploit availability
2. Lack of correlation between vulnerabilities and exploits
3. Time-consuming manual research

3.3.2 Solution:

1. Use Searchsploit to map vulnerabilities to exploits
2. Extract exploit descriptions and references
3. Prioritize vulnerabilities based on exploit availability

3.3.3 Advantages:

1. Saves time in vulnerability analysis
2. Provides real-world attack insights
3. Improves risk prioritization
4. Enhances penetration testing workflow

3.4 PRIVILEGE ESCALATION DETECTION USING AUTOMATED TOOLS

Authors: Security researchers

Published: 2022

Overview:

This study focuses on identifying privilege escalation opportunities in systems using tools like LinPEAS. It analyzes system configurations, permissions, and vulnerabilities to detect possible escalation paths.

3.4.1 Problem:

1. Privilege escalation paths are difficult to identify manually
2. Misconfigurations often go unnoticed
3. Lack of automated analysis

3.4.2 Solution:

1. Use automated tools to scan system configurations
2. Identify weak permissions and SUID binaries
3. Detect misconfigurations and vulnerabilities

3.4.3 Advantages:

1. Improves system security visibility
2. Detects hidden vulnerabilities
3. Supports post-exploitation analysis
4. Reduces manual effort

3.5 GRAPH BASED SECURITY ANALYSIS TOOL USING NEO4J

Authors: Data security researchers

Published: 2023

Overview:

This paper discusses using graph databases like Neo4j for modeling and analyzing cybersecurity data. It enables visualization of relationships between assets, vulnerabilities, and attack paths.

3.5.1 Problem:

1. Traditional databases cannot represent complex relationships effectively
2. Difficulty in analyzing interconnected vulnerabilities
3. Lack of visualization tools

3.5.2 Solution:

1. Use Neo4j to store nodes and relationships
2. Represent systems, vulnerabilities, and exploits as graph elements
3. Query attack paths using Cypher

3.5.3 Advantages:

1. Efficient representation of relationships
2. Enables attack path visualization
3. Supports advanced queries
4. Improves analysis capabilities

3.6 SUMMARY OF LITERATURE SURVEY

From the above studies, it is clear that:

- Existing systems focus on **individual tasks** like scanning, vulnerability detection, or exploit mapping
- There is a lack of **integrated frameworks** that combine all these processes
- Most systems do not provide **complete attack path visualization**

3.6.1 Research Gap Identified:

1. Lack of unified system combining scanning, vulnerability mapping, and attack path generation
2. Limited automation in correlating multiple tools
3. Absence of real-time attack path visualization

3.6.2 Proposed Contribution:

The proposed **Attack Path Discovery Framework** addresses these gaps by:

- Integrating tools like Nmap, OpenVAS, Searchsploit, LinPEAS, and BloodHound
- Automating vulnerability correlation
- Generating realistic attack paths
- Visualizing attack graphs using Neo4j
- Ensuring safe analysis without exploitation

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 SYSTEM REQUIREMENT SPECIFICATION (SRS)

The Attack Path Discovery Framework is designed to automate vulnerability assessment and generate realistic attack paths by integrating multiple cybersecurity tools such as Nmap, OpenVAS, Searchsploit, LinPEAS, and BloodHound. The system collects target input (IP address or domain), performs safe scanning, analyzes vulnerabilities, correlates data, and visualizes attack paths using a graph database.

The system architecture is built using Node.js (backend), React.js (frontend), and Neo4j (graph database) to ensure scalability, modularity, and efficient visualization of attack relationships.

Functionally, the system performs:

- Network scanning and service detection
- Vulnerability mapping and analysis
- Exploit intelligence gathering (read-only)
- Privilege escalation simulation
- Attack path correlation
- Graph-based visualization of attack paths

The system is designed to ensure safe and ethical usage, meaning it does not perform actual exploitation but only simulates possible attack scenarios based on available data.

4.1.1 Core Functional Features

1. Target Input Module

- Accepts IP address or domain name as input
- Validates and preprocesses target information

2. Network Scanning Module

- Uses Nmap to identify open ports, services, and OS details
- Performs safe and non-intrusive scanning

3. Vulnerability Assessment Module

- Uses OpenVAS or CVE database
- Identifies known vulnerabilities associated with services

4. Exploit Mapping Module

- Uses Searchsploit to map vulnerabilities with available exploits
- Works in read-only mode (no execution)

5. Privilege Escalation Analysis Module

- Simulates outputs similar to LinPEAS
- Identifies possible privilege escalation paths

6. Active Directory Analysis Module (Optional)

- Uses BloodHound data for AD environments
- Identifies lateral movement paths

7. Correlation Engine

- Combines data from all modules
- Generates multi-step attack paths

8. Attack Graph Generation Module

- Creates nodes and relationships
- Stores data in Neo4

9. Visualization Module

- Displays attack graph using React and Neo4j
- Provides interactive analysis

10. Reporting Module

- Generates structured JSON reports
- Displays attack paths, severity, and recommendations

4.2 HARDWARE REQUIREMENTS

The system requires moderate hardware for smooth operation:

1. **Processor:** Intel Core i5 or higher
2. **RAM:** Minimum 8 GB (16 GB recommended)
3. **Hard Disk:** 500 GB or more
4. **Network:** Stable internet connection for tool integration
5. **Optional GPU:** For faster data processing (not mandatory)

4.3 SOFTWARE REQUIREMENTS

1. Operating System

- Windows 10 / Linux / macOS

2. Programming Languages

- JavaScript (Node.js for backend)
- JavaScript (React.js for frontend)

3. Backend Framework

- Node.js with Express.js

4. Frontend Framework

- React.js

5. Database

- Neo4j (Graph Database)

6. Cybersecurity Tools

- Nmap → Network scanning
- OpenVAS → Vulnerability assessment
- Searchsploit → Exploit mapping

- LinPEAS → Privilege escalation analysis
- BloodHound → Active Directory analysis

7. Libraries & Tools

- neo4j-driver → Database connectivity
- axios / fetch → API calls
- child_process → Tool execution
- Cytoscape.js / D3.js → Graph visualization

4.4 SOFTWARE DESCRIPTION

The **Attack Path Discovery Framework** is an intelligent cybersecurity system that integrates multiple tools and technologies to analyze vulnerabilities and generate realistic attack paths without performing actual exploitation.

4.4.1 Backend System (Node.js)

The backend is implemented using Node.js, which acts as the core engine responsible for:

- Executing security tools using child processes
- Parsing tool outputs
- Correlating vulnerability data
- Generating attack paths

The backend follows a modular structure:

- Scanner module
- Vulnerability analysis module
- Correlation engine

4.4.2 Frontend System (React.js)

The frontend provides an interactive dashboard where users can:

- Enter target IP or domain

- View scan results
- Analyze vulnerabilities
- Visualize attack paths

The interface is designed to be:

- User-friendly
- Responsive
- Real-time interactive

4.4.3 Graph Database (Neo4j)

Neo4j is used to store and visualize relationships between:

- Targets
- Services
- Vulnerabilities
- Exploits
- Attack steps

Each element is represented as a node, and relationships define how an attacker can move through the system.

4.4.4 Attack Path Generation

The system builds attack paths by correlating:

- Open ports → Services
- Services → Vulnerabilities
- Vulnerabilities → Exploits
- Exploits → Access levels

Example:

Port 80 → Apache Service → CVE → Exploit Available → Initial Access → Privilege Escalation

4.4.5 Visualization System

The visualization module uses Neo4j and React-based graph libraries to:

- Display attack paths
- Show relationships between vulnerabilities
- Allow interactive exploration

4.4.6 Security and Ethical Considerations

The system is designed with strict safety measures:

- No exploit execution
- Only analysis and simulation
- Works in controlled environments
- Suitable for educational use

4.4.7 System Advantages

- Automates vulnerability assessment
- Provides real-world attack path analysis
- Enhances understanding of security risks
- Improves decision-making for security analysts
- Offers interactive visualization

CHAPTER 5

SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

“Attack Path Discovery and Severity Assessment Framework for Offensive Security Analysis”

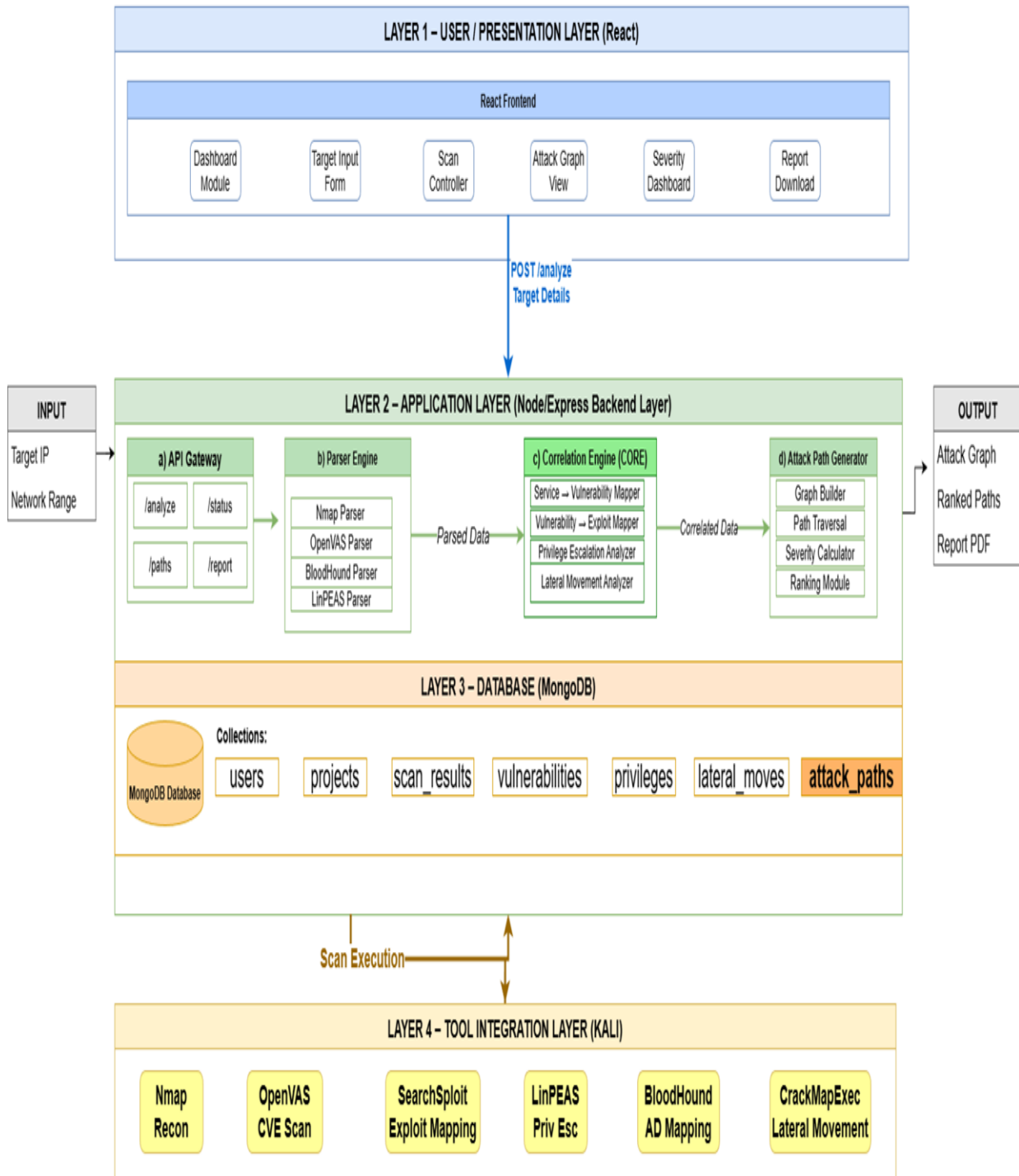


Fig 5.1.1 System Architecture

5.2 ER DIAGRAM

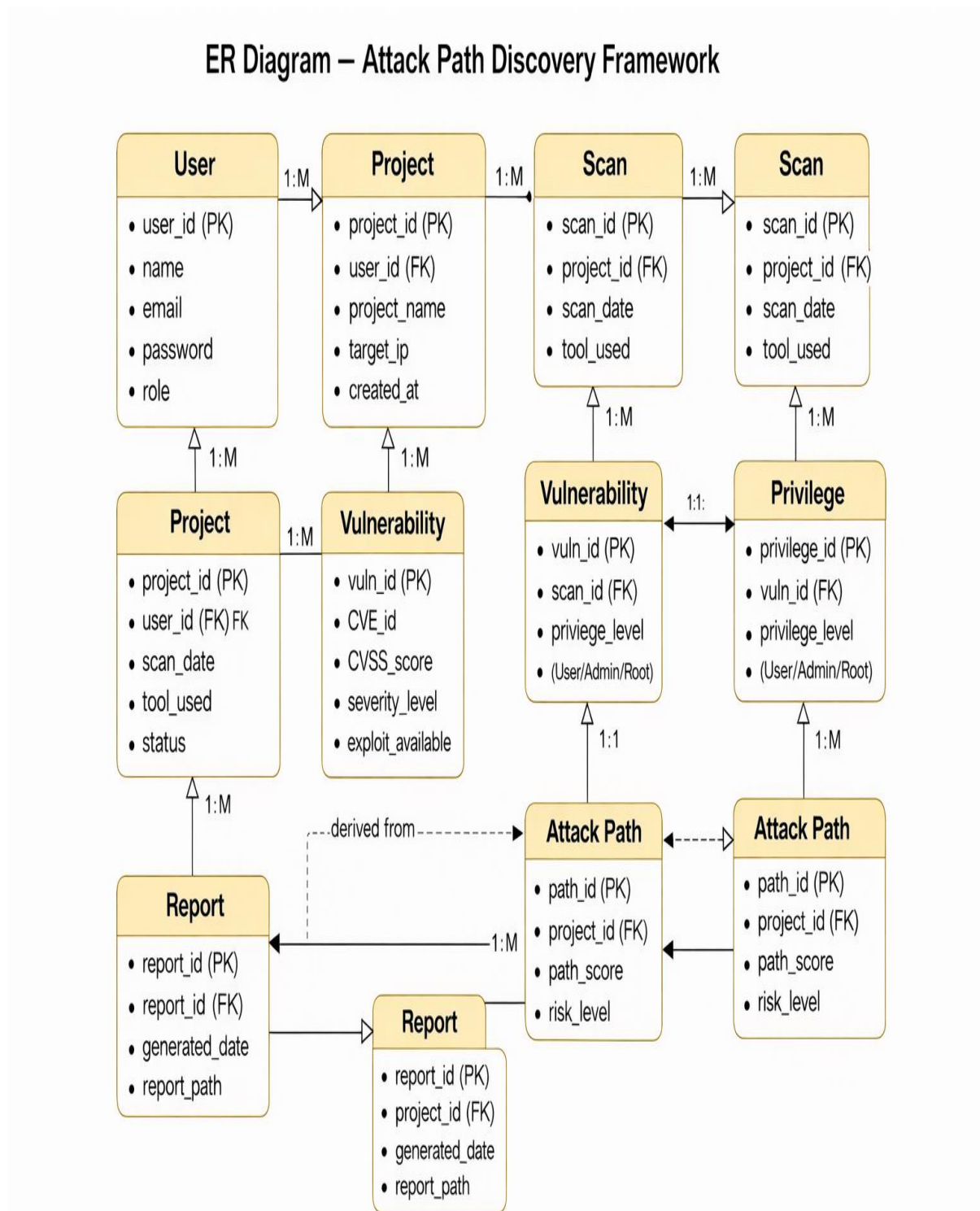


Fig 5.2.1 ER Diagram

5.3 DATA FLOWDIAGRAM

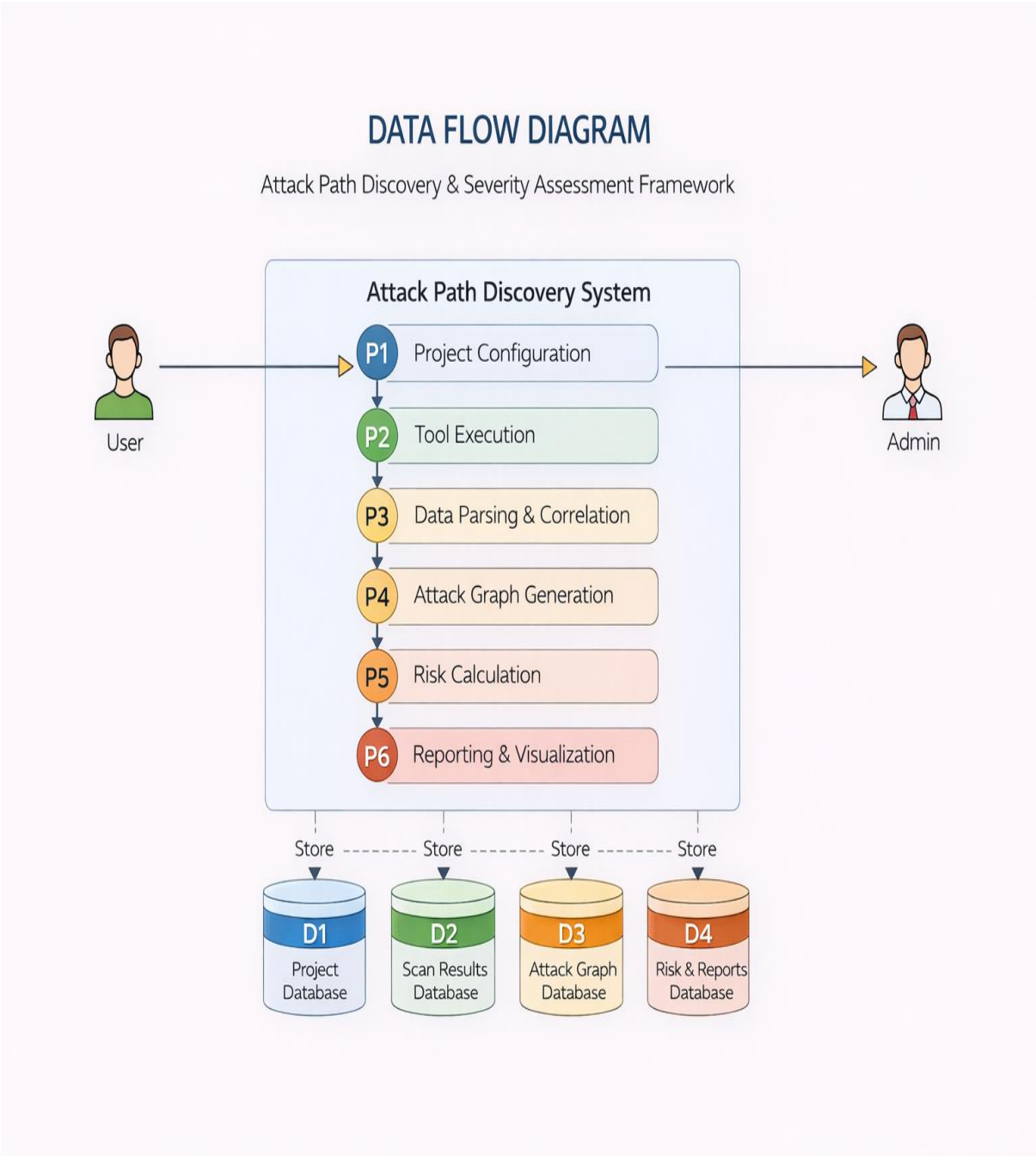


Fig 5.3.1 Data Flow Diagram

5.4 USE CASE DIAGRAM

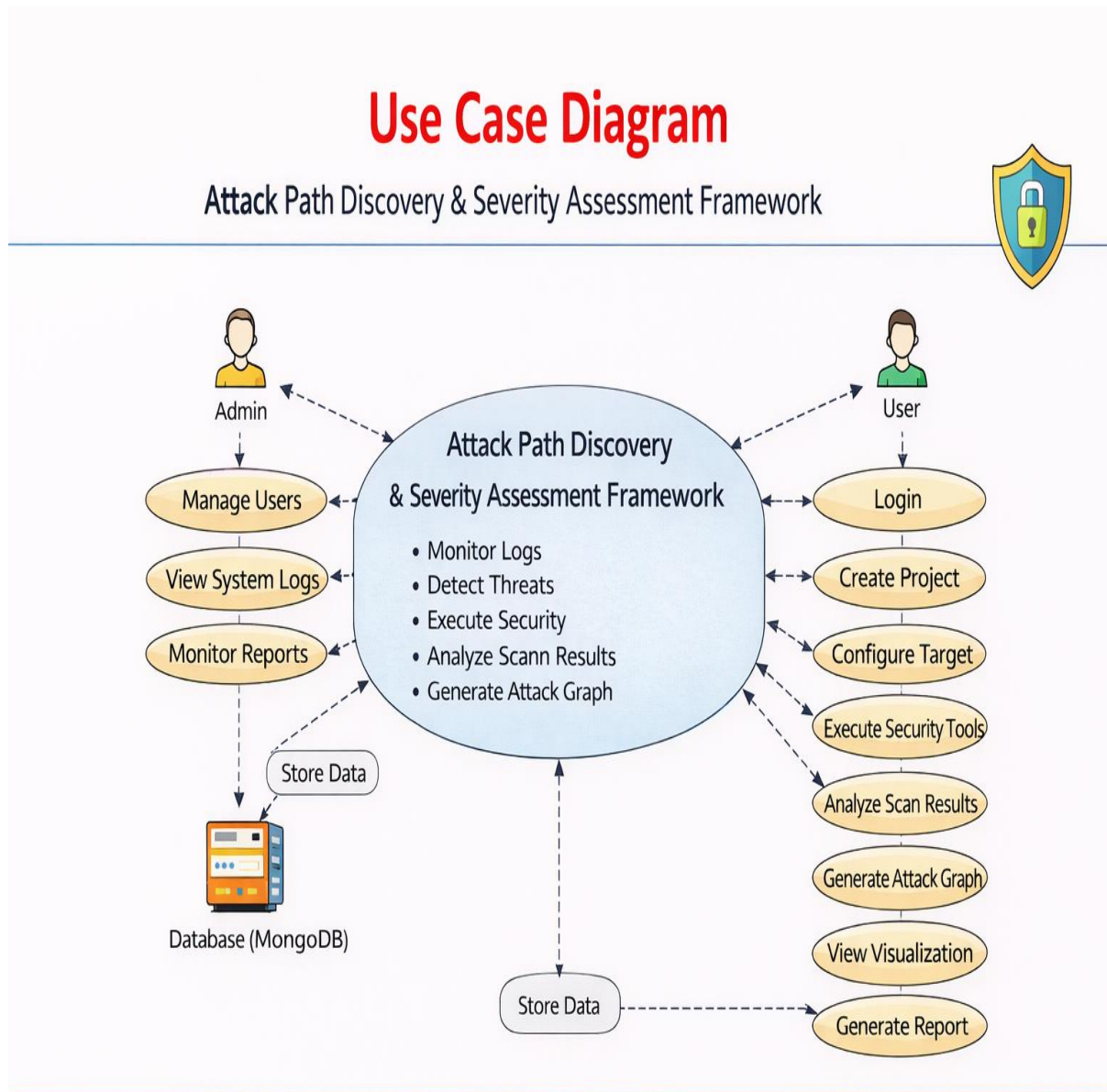


Fig 5.4.1 Use Case Diagram

5.5 ARCHITECTURE DIAGRAM EXPLANATION

- The system architecture of the *Attack Path Discovery and Severity Assessment Framework* follows a modular and scalable design using the MERN stack integrated with cybersecurity tools and a graph database.
- The architecture consists of three main layers: Frontend Layer, Backend Layer, and Data Layer.
- The Frontend Layer (React.js) provides an interactive user interface where users can input target IP addresses or domain names, initiate scans, and visualize results. It displays vulnerabilities, attack paths, and graph-based relationships using visualization libraries like Cytoscape.js.
- The Backend Layer (Node.js with Express.js) acts as the core processing unit. It handles API requests, executes security tools such as Nmap, OpenVAS, and Searchsploit using child processes, and performs vulnerability analysis. The backend also includes a correlation engine that links services, vulnerabilities, exploits, and privilege escalation paths to generate multi-step attack paths.
- The Data Layer consists of a Neo4j graph database and structured JSON storage. Neo4j stores nodes such as systems, services, vulnerabilities, and exploits, along with their relationships, enabling efficient attack path querying and visualization.
- The architecture ensures secure communication using JWT authentication, supports scalability, and enables real-time analysis and visualization of cyber threats.

5.6 USE CASE DIAGRAM EXPLANATION

The use case diagram represents the interaction between users and the Attack Path Discovery System.

The primary actor in the system is the User (Security Analyst / Ethical Hacker).

The user interacts with the system through the following use cases:

- Login/Register: The user securely accesses the system using authentication mechanisms.
- Upload Target/Input IP: The user provides targets such as IP addresses or domain names.
- Start Scan: Initiates vulnerability scanning and analysis.
- View Scan Results: Displays detected vulnerabilities and system information.
- Analyze Attack Paths: Allows the user to explore generated attack paths.

- View Graph Visualization: Displays attack paths in graphical format for better understanding.
- Filter by Severity: Enables prioritization of vulnerabilities based on risk levels.
- View Scan History: Allows users to access previous scan records.

The system processes these actions internally by performing scanning, analysis, correlation, and visualization, and returns meaningful insights to the user.

5.7 DATA FLOW DIAGRAM EXPLANATION

- The data flow in the system represents how information moves through different components of the Attack Path Discovery Framework.
- Initially, the user provides input in the form of a target IP address or domain through the frontend interface. This input is sent to the backend via API requests.
- The backend triggers the network scanning module (Nmap), which identifies open ports, services, and system details. The output is then passed to the vulnerability analysis module, where services are mapped to known vulnerabilities using CVE databases.
- Next, the system performs exploit mapping using Searchsploit, identifying available exploits related to the detected vulnerabilities. Simultaneously, the privilege escalation module analyzes system configurations to detect possible escalation paths.
- All collected data is then processed by the correlation engine, which links ports, services, vulnerabilities, exploits, and access levels to form attack chains.
- The generated attack paths are stored in the Neo4j graph database, where nodes and relationships represent the attack structure.
- Finally, the processed data is sent back to the frontend, where it is displayed as visual graphs, reports, and risk insights, enabling users to understand and analyze potential attack paths effectively.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 NETWORK SCANNING USING NMAP

The first stage of the system implementation involves gathering information about the target system using network scanning techniques. The system integrates Nmap (Network Mapper) to perform safe and non-intrusive scanning.

6.1.1 Scanning Process

- The user provides a target IP address or domain name.
- The system executes Nmap using Node.js `child_process`.
- The command used: `nmap -sC -sV -A -T4 <target>`

6.1.2 Output Extraction

The system parses Nmap results to extract:

- Open ports
- Running services
- Service versions
- Operating system details

6.1.3 Purpose

- Identify attack surface
- Detect entry points
- Provide input for further analysis

6.2 VULNERABILITY ANALYSIS

After scanning, the system performs vulnerability mapping based on detected services.

6.2.1. Vulnerability Detection

Matches services and versions with known vulnerabilities

Uses:

- OpenVAS (if integrated)
- Local CVE database

6.2.2 Output Data

- CVE ID
- Vulnerability description
- Severity level (CVSS score)

6.2.3 Implementation

- Service → CVE mapping is performed using predefined datasets or APIs
- Results are stored in structured JSON format

6.2.4 Purpose

- Identify security weaknesses
- Provide basis for attack path generation

6.3 EXPLOIT MAPPING USING SEARCHSPLOIT

The system uses Searchsploit to identify available exploits for detected vulnerabilities.

6.3.1 Process

- Extract service version from Nmap results
- Execute:

```
searchsploit <service version>
```

6.3.2 Data Extracted

- Exploit title
- Exploit path
- Reference links

6.3.3 Important Note

- Exploits are **NOT** executed
- Only analyzed for availability

6.3.4 Purpose

- Determine exploitability
- Enhance risk analysis

6.4 PRIVILEGE ESCALATION ANALYSIS

The system simulates privilege escalation possibilities using logic inspired by LinPEAS.

6.4.1 Analysis Includes

- SUID binaries
- Weak file permissions
- Cron jobs
- Misconfigurations

6.4.2 Implementation

- Uses predefined rules and simulated outputs
- Detects possible escalation paths

6.4.3 Output

- Potential privilege escalation vectors
- Risk level

6.4.4 Purpose

- Identify post-exploitation opportunities
- Extend attack paths

6.5 ACTIVE DIRECTORY ANALYSIS

For enterprise environments, the system supports Active Directory analysis.

6.5.1 Tool Used

- BloodHound

6.5.2 Functionality

- Identifies user relationships
- Detects privilege chains
- Maps lateral movement paths

6.5.3 Output

- Graph-based relationships
- Attack paths in AD environment

6.6 CORRELATION ENGINE

The correlation engine is the core component of the system.

6.6.1 Function

Combines data from:

- Nmap
- Vulnerability scanner
- Searchsploit
- Privilege analysis

6.6.2 Logic

Maps:

- Port → Service
- Service → Vulnerability
- Vulnerability → Exploit
- Exploit → Access

6.6.3 Example Attack Chain

Port 22 → SSH Service → Vulnerability → Exploit Available → Access → Privilege Escalation

6.6.4 Output

- Multi-step attack paths
- Ranked based on severity

6.7 ATTACK PATH GENERATION

The system generates realistic attack paths using correlated data.

1. Structure of Each attack path contains:

- Step-by-step actions
- Tools involved
- Expected outcomes

2. Ranking Criteria

- Severity (CVSS score)
- Exploit availability

- Complexity

3. Output Format

```
{
  "path_id": "1",
  "steps": [
    {
      "stage": "Initial Access",
      "action": "Exploit vulnerability",
      "tool": "Searchsploit",
      "confidence": "High"
    }
  ],
  "risk_score": "High"
}
```

6.8 GRAPH DATABASE IMPLEMENTATION (NEO4J)

The system uses Neo4j to store and represent attack paths.

1. Node Types

- Target
- Port
- Service
- Vulnerability
- Exploit
- Access Level

2. Relationships

- RUNS_ON
- HAS_VULNERABILITY
- CAN_EXPLOIT
- LEADS_TO
- ESCALATES_TO

3. Example Query

```

CREATE                                (t:Target                                {ip:"192.168.1.1"})
CREATE                                (s:Service                                {name:"Apache"})
CREATE                                (v:Vulnerability                        {cve:"CVE-XXXX"})
CREATE                                (t)-[:RUNS_ON]->(s)
CREATE (s)-[:HAS_VULNERABILITY]->(v)

```

4. Purpose

- Store relationships efficiently
- Enable attack path queries

6.9 FRONTEND IMPLEMENTATION (REACT)

The frontend is developed using React.js.

1. Features

- Input field for target
- Scan progress display
- Result dashboard

2. Display Modules

- Open ports
- Vulnerabilities
- Attack paths
- Graph visualization

3. Visualization

- Uses Cytoscape.js or D3.js
- Displays interactive graphs

6.10 API IMPLEMENTATION (NODE.JS)

The backend provides REST APIs using Express.js.

1. APIs

- POST /scan → Starts scanning
- GET /results → Returns analysis
- GET /graph → Returns Neo4j data

2. Function

- Handles tool execution
- Processes data
- Sends structured output

6.11 DATA VISUALIZATION

The system provides visual representation of attack paths.

1. Graph Visualization

- Nodes → System components
- Edges → Attack steps

2. Charts

- Risk levels
- Vulnerability distribution

3. Purpose

- Improve understanding
- Support decision-making

6.12 SYSTEM WORKFLOW

The overall workflow of the system:

1. User enters target IP/domain
2. Nmap scans the target
3. Vulnerabilities are identified
4. Exploits are mapped
5. Privilege escalation is analyzed
6. Data is correlated
7. Attack paths are generated
8. Results are visualized

6.13 SECURITY AND ETHICAL IMPLEMENTATION

The system follows strict ethical guidelines:

- No real exploitation performed
- Only simulation and analysis
- Works in controlled environments
- Designed for educational use\

CHAPTER 7

METHODOLOGY

7.1 DATA COLLECTION AND PREPROCESSING

7.1.1 Data Collection Overview

The Attack Path Discovery Framework collects data from multiple sources to analyze vulnerabilities and generate attack paths. The primary data sources include:

- Network scanning results (Nmap)
- Vulnerability databases (CVE, OpenVAS reports)
- Exploit databases (Searchsploit)
- System configuration data (for privilege escalation analysis)

The collected data is mostly unstructured and requires preprocessing before analysis.

7.1.2 Preprocessing Objectives

The preprocessing stage ensures that the collected data is clean, structured, and ready for correlation and analysis. The main objectives are:

- Normalize raw scan outputs
- Extract meaningful information from unstructured data
- Remove redundant or irrelevant data
- Convert data into structured formats (JSON)
- Ensure compatibility with analysis modules

7.1.3 Preprocessing Steps

a. Data Cleaning

- Remove unnecessary symbols, logs, and noise from tool outputs
- Filter only relevant information such as ports, services, and vulnerabilities

b. Text Normalization

- Convert all extracted data into a consistent format
- Standardize service names and versions
- Handle variations in vulnerability naming

c. Feature Extraction

- Extract key attributes:
 - Port number
 - Service name
 - Version
 - CVE IDs
- Store extracted data in structured format

d. Data Structuring

- Convert extracted data into JSON format
- Organize into categories:
 - Network data
 - Vulnerability data
 - Exploit data

7.1.4 Outcome of Preprocessing

After preprocessing, the data becomes:

- Clean and structured
- Ready for correlation
- Suitable for attack path generation
- Compatible with graph database storage

7.2 NETWORK SCANNING AND VULNERABILITY IDENTIFICATION

7.2.1 Network Scanning

The system uses Nmap to scan the target system and identify:

- Open ports
- Running services
- Service versions
- Operating system details

This step provides the foundation for further analysis.

7.2.2 Vulnerability Identification

- The detected services are mapped with known vulnerabilities
- CVE databases are used to identify vulnerabilities
- Each vulnerability is assigned a severity score

7.2.3 Output

- List of open ports
- Identified services
- Associated vulnerabilities

7.3 EXPLOIT ANALYSIS AND MAPPING

Exploit Identification

The system uses Searchsploit to identify available exploits for detected vulnerabilities.

Process

- Match service versions with exploit database
- Extract exploit information:
 - Title
 - Description
 - Reference links

Important Note

- Exploits are not executed
- Only analyzed for availability

Purpose

- Determine exploit feasibility
- Enhance risk analysis

7.4 PRIVILEGE ESCALATION ANALYSIS

Objective

To identify possible ways an attacker can gain higher privileges after initial access.

7.4.2. Methodology

Analyze system configurations

Identify:

- Weak permissions
- SUID binaries
- Misconfigurations

7.5 Output

- List of possible escalation paths
- Risk level of each path

7.5 ATTACK PATH CORRELATION

Correlation Process

The system combines all collected data to generate attack paths.

Mapping Logic

- Port → Service
- Service → Vulnerability
- Vulnerability → Exploit
- Exploit → Access
- Access → Privilege escalation

Example

Port 80 → Web Server → Vulnerability → Exploit → Shell Access → Root Access

Purpose

- Identify multi-step attack chains
- Provide realistic attack scenarios

7.6 ATTACK PATH GENERATION

Process

- Generate multiple attack paths
- Each path consists of sequential steps

Path Structure

Each attack path includes:

- Stage (Reconnaissance, Access, Escalation)
- Action performed
- Tools used
- Result

Ranking Criteria

Attack paths are ranked based on:

- Severity of vulnerabilities
- Exploit availability
- Complexity of attack

Output

- Multiple attack paths
- Risk scores
- Confidence levels

7.7 GRAPH-BASED VISUALIZATION

Objective

To visually represent attack paths for better understanding.

Implementation

Use Neo4j graph database

Represent:

- Nodes → System components
- Edges → Attack steps

Visualization Tools

- React-based graph libraries
- Cytoscape.js / D3.js

Benefits

- Easy understanding of attack flow

- Interactive exploration
- Improved decision-making

7.8 SYSTEM WORKFLOW

The complete workflow of the system is as follows:

1. User provides target IP/domain
2. Nmap performs network scanning
3. Vulnerabilities are identified
4. Exploits are mapped
5. Privilege escalation is analyzed
6. Data is correlated
7. Attack paths are generated
8. Results are visualized

7.9 SAFETY AND ETHICAL CONSIDERATIONS

The system is designed to ensure safe and ethical usage:

- No real exploitation is performed
- Only simulation and analysis
- Works in authorized environments
- Designed for educational purposes

7.10 SUMMARY OF METHODOLOGY

The methodology integrates multiple stages:

- Data collection
- Preprocessing
- Vulnerability analysis
- Exploit mapping
- Attack path generation
- Visualization

This structured approach ensures accurate and meaningful security analysis while maintaining ethical standards

CHAPTER 8

SAMPLE CODING

8.1 SAMPLE CODING

Scan Orchestrator — Pipeline coordinator

```
// REAL EXECUTION ONLY. NO MOCK DATA.
```

```
const { v4: uuidv4 } = require('uuid');
```

```
const { runNmapScan } = require('./engines/recon');
```

```
const { mapVulnerabilities } = require('./engines/vulnMapper');
```

```
const { searchExploits } = require('./engines/exploitIntel');
```

```
const { simulatePrivEsc } = require('./engines/privEscSimulator');
```

```
const { analyzeAD } = require('./engines/adAnalyzer');
```

```
const { correlate } = require('./engines/correlationEngine');
```

```
const { generatePaths } = require('./engines/pathGenerator');
```

```
// In-memory store for scan results
```

```
const scanStore = new Map();
```

```
/**
```

```
 * Run the full scan pipeline for a target.
```

```
 * Every step uses REAL tool execution. No mock data. No fallbacks.
```

```
 *
```

```
 * @param {string} target - IP or domain to scan
```

```
 * @param {function} emit - Socket.IO emit function for progress updates
```

```
 * @returns {Object} full scan results
```

```
 */
```

```
async function runScan(target, emit = () => {}) {
```

```
  const scanId = uuidv4();
```

```
  const startTime = Date.now();
```

```

console.log(\n${'='.repeat(60)});
console.log(` SCAN STARTED: ${target} [${scanId}]`);
console.log('${'='.repeat(60)}\n');
const progress = (phase, pct, message) => {
  emit('scan:progress', { scanId, phase, progress: pct, message });
  console.log(` [${pct}%] ${phase}: ${message}`);
};
try {
  // — Phase 1: Reconnaissance (REAL Nmap) —
  progress('Reconnaissance', 10, 'Running Nmap...');
  const scanResult = await runNmapScan(target);
  progress('Reconnaissance', 25, Found ${scanResult.ports.length} open ports);
  // — Phase 2: Vulnerability Mapping (CVE database lookup) —
  progress('Vulnerability Mapping', 30, 'Mapping services to known CVEs...');
  const vulnerabilities = await mapVulnerabilities(scanResult.ports);
  progress('Vulnerability Mapping', 45, Found ${vulnerabilities.length}
vulnerabilities);

  // — Phase 3: Exploit Intelligence (REAL Searchsploit) —
  progress('Exploit Intelligence', 50, 'Running Searchsploit...');
  const exploits = await searchExploits(scanResult.ports);
  progress('Exploit Intelligence', 60, Found ${exploits.length} potential exploits);
  // — Phase 4: Privilege Escalation Analysis —
  progress('Privilege Escalation', 65, 'Analyzing privilege escalation vectors...');
  const privEscVectors = await simulatePrivEsc(scanResult);
  progress('Privilege Escalation', 75, Identified ${privEscVectors.length} vectors);
  // — Phase 5: Active Directory Analysis —
  progress('Active Directory', 78, 'Checking for AD indicators...');
  const adAnalysis = await analyzeAD(scanResult);

```

```
progress('Active Directory', 82, adAnalysis ? 'AD analysis complete' : 'No AD environment detected');
```

```
// — Phase 6: Correlation (builds graph from REAL data) —
```

```
progress('Correlation', 85, 'Correlating findings into attack chains...');
```

```
// FIX: Pass data with the correct keys that correlate() expects
```

```
const correlationInput = {
```

```
  recon: scanResult,
```

```
  vulns: vulnerabilities,
```

```
  exploits,
```

```
  privEsc: privEscVectors,
```

```
  ad: adAnalysis,
```

```
};
```

```
const graph = correlate(correlationInput);
```

```
progress('Correlation', 90, Built graph: ${graph.nodes.length} nodes,  
${graph.edges.length} edges);
```

```
// — Phase 7: Attack Path Generation (data-driven ONLY) —
```

```
progress('Path Generation', 92, 'Generating attack paths...');
```

```
// FIX: Pass (graph, scanId) — the correct signature for generatePaths
```

```
const attackPaths = await generatePaths(graph, scanId);
```

```
progress('Path Generation', 98, Generated ${attackPaths.length} attack paths);
```

```
// — Build final result (flat format per spec) —
```

```
const result = {
```

```
  scanId,
```

```
  target: scanResult.target,
```

```
  scanTime: scanResult.scanTime,
```

```
  duration: ${((Date.now() - startTime) / 1000).toFixed(1)}s,
```

```
  os: scanResult.os,
```

```
  ports: scanResult.ports.map(p => p.number),
```

```

services: scanResult.ports.map(p => ({
  port: p.number,
  name: p.service.name,
  version: ${p.service.product || ""} ${p.service.version || ""}.trim() || 'unknown',
})),
vulnerabilities,
exploits,
privEscVectors,
adAnalysis,
graph,
attackPaths,
summary: {
  totalPorts: scanResult.ports.length,
  totalVulnerabilities: vulnerabilities.length,
  totalExploits: exploits.length,
  totalPrivEscVectors: privEscVectors.length,
  totalAttackPaths: attackPaths.length,
  highestRisk: attackPaths.length > 0
    ? Math.max(...attackPaths.map(p => p.riskScore || 0))
    : 0,
  overallSeverity: getOverallSeverity(attackPaths),
},
status: 'success',
timestamp: new Date().toISOString(),
};
// Store result in memory
scanStore.set(scanId, result);

```

```

    progress('Complete', 100, 'Scan complete!');
    emit('scan:complete', { scanId });
    console.log(\n${'='.repeat(60)});
    console.log(`  SCAN COMPLETE: ${attackPaths.length} paths, highest risk:
${result.summary.highestRisk}`);
    console.log(`${'='.repeat(60)}\n`);
    return result;
  } catch (err) {
    console.error([Orchestrator] Scan failed: ${err.message});
    emit('scan:error', { scanId, error: err.message });
    throw err;
  }
}

function getOverallSeverity(paths) {
  if (paths.length === 0) return 'Info';
  const highest = Math.max(...paths.map(p => p.riskScore || 0));
  if (highest >= 9) return 'Critical';
  if (highest >= 7) return 'High';
  if (highest >= 4) return 'Medium';
  return 'Low';
}

/**
 * Get stored scan result by ID.
 */

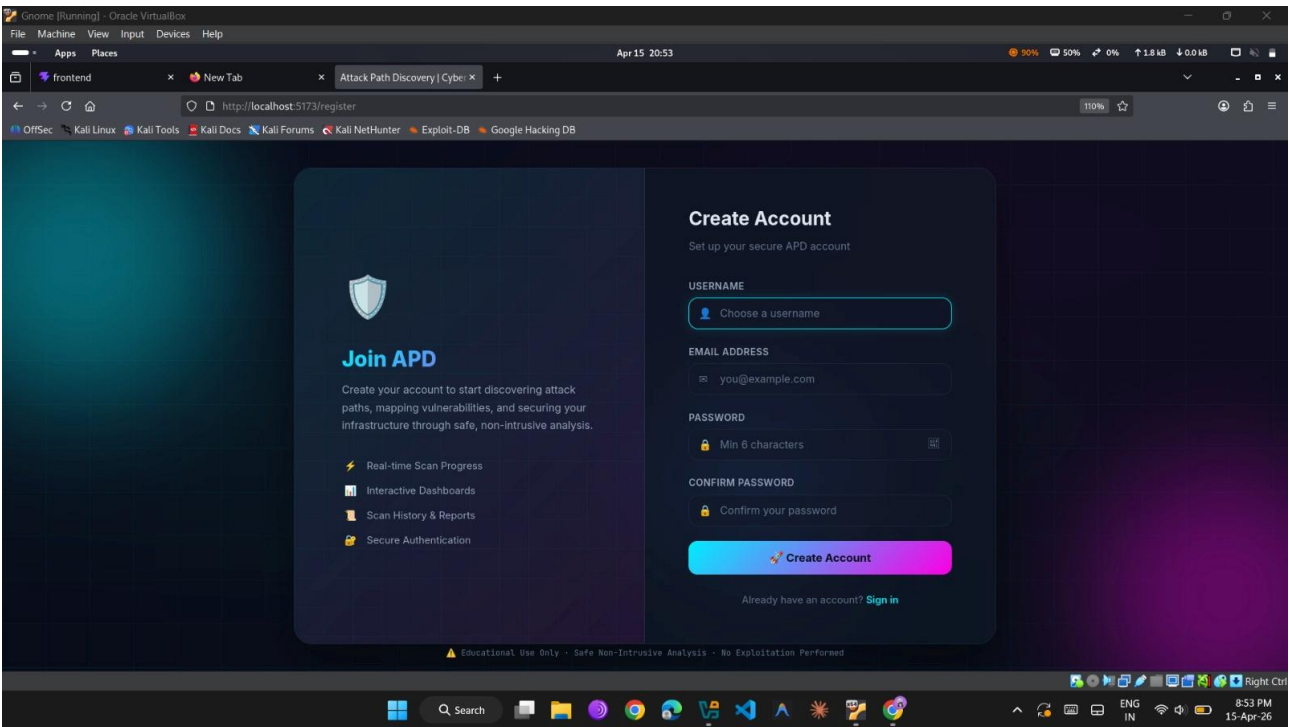
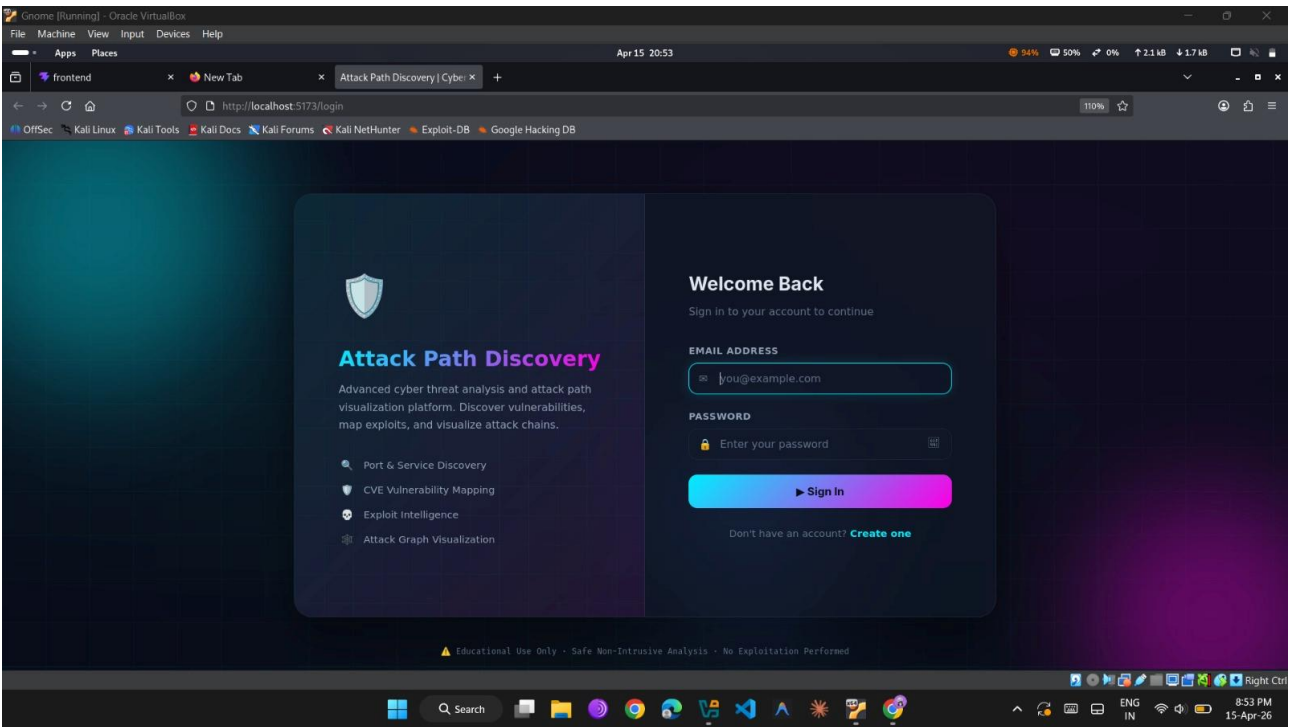
function getScan(scanId) {
  return scanStore.get(scanId);
}

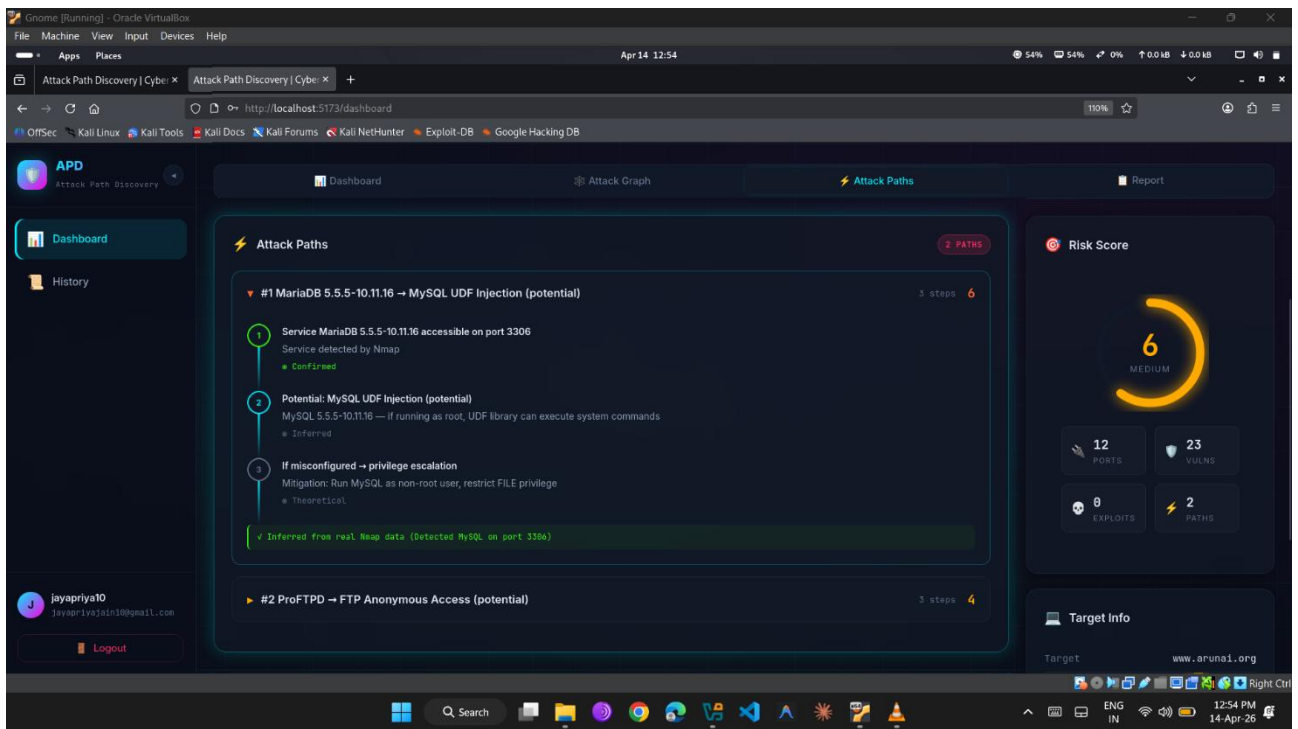
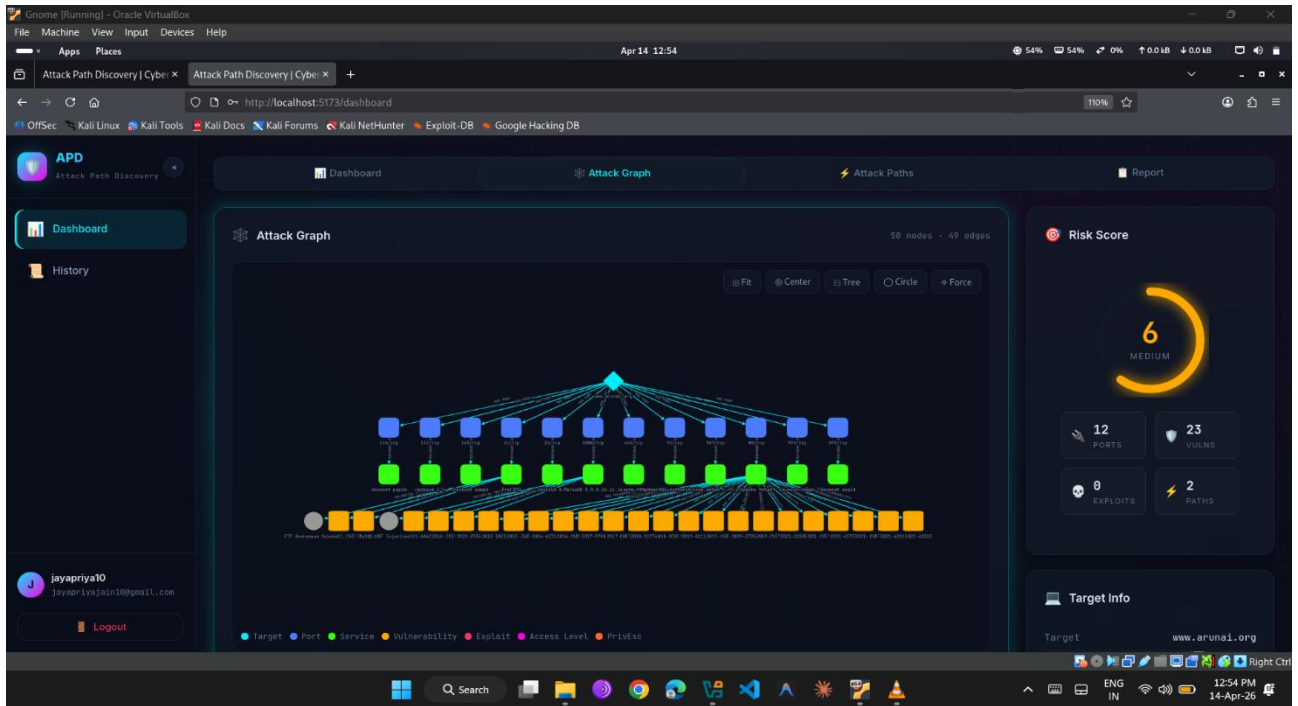
/**

```

```
* Get the latest scan result.
*/
function getLatestScan() {
  const entries = Array.from(scanStore.entries());
  if (entries.length === 0) return null;
  return entries[entries.length - 1][1];
}
/**
 * Get all scan IDs.
 */
function getAllScanIds() {
  return Array.from(scanStore.keys());
}
module.exports = { runScan, getScan, getLatestScan, getAllScanIds };
```

8.2 SCREENSHOTS





GNOME [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Apr 14 12:55

77% 55% 0% 0.1 KB 17 KB

Attack Path Discovery | Cyber

http://localhost:5173/dashboard

100%

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

APD Attack Path Discovery

Dashboard Attack Graph Attack Paths Report

JSON Report

Copy JSON Download

```
{
  "target": "www.arunai.org",
  "scanId": "9e51688e-cf0d-489d-8d9b-486a281a17b5",
  "duration": "42.6s",
  "ports": [
    21,
    22,
    53,
    80,
    110,
    111,
    143,
    443,
    587,
    993,
    995,
    3306
  ],
  "services": [
    {
      "port": 21,
      "name": "ftp",

```

Risk Score

6 MEDIUM

12 PORTS 23 VULNS

0 EXPLOITS 2 PATHS

Target Info

Target www.arunai.org

jayapriya10 jayapriya10@gmail.com Logout

Search

12:55 PM 14-Apr-26

GNOME [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Apr 13 15:25

36% 63% 2% 0.0 KB 0.0 KB

Attack Path Discovery | Cyber

http://localhost:3000/dashboard

100%

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

APD Attack Path Discovery

Dashboard Attack Graph Attack Paths Report

Ports 23 Services 23 Vulnerabilities 43 Exploits 0 PrivEsc 4

PORT	STATE
21	open
22	open
23	open
28	open
53	open
80	open
111	open
139	open
443	open
512	open
513	open
514	open
1099	open

Risk Score

6 MEDIUM

23 PORTS 43 VULNS

0 EXPLOITS 3 PATHS

Target Info

Target 10.140.37.146

OS Unknown

Family Unknown

Accuracy 0%

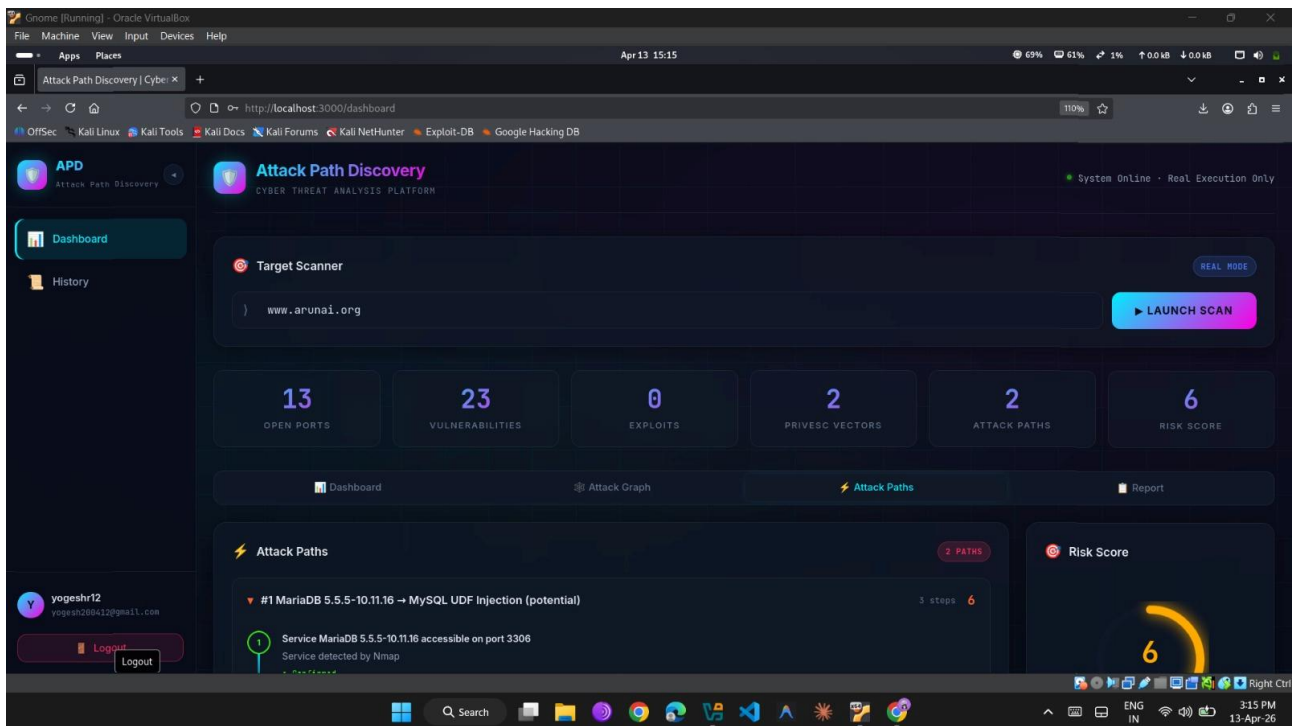
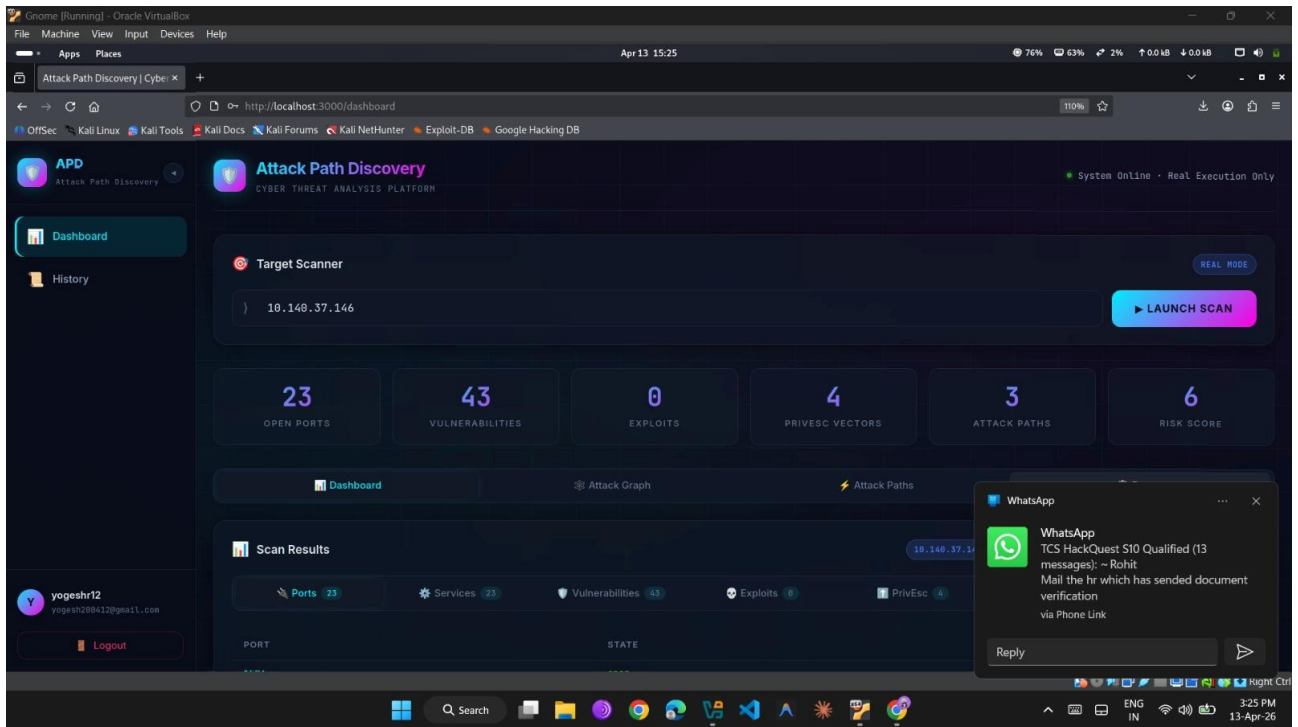
Scan Time 33.3s

Scan ID 9849a5f1...

yogesh12 yogesh209412@gmail.com Logout

Search

3:25 PM 13-Apr-26



CHAPTER 9

TESTING AND VALIDATION

9.1 INTRODUCTION TO TESTING

Testing is a crucial phase in the development of the Attack Path Discovery Framework to ensure that all modules function correctly and meet the required objectives. It helps in verifying the performance, reliability, and accuracy of the system in detecting vulnerabilities and generating attack paths.

The testing process involves evaluating both individual components and the complete system under different scenarios.

The system is tested using both normal and simulated attack inputs to ensure that it can accurately identify security risks and vulnerabilities.

Various scanning and detection processes are validated to check the effectiveness of the system. This ensures that the framework performs reliably and produces accurate results in real-time environments.

9.2 TYPES OF TESTING

9.2.1 Unit Testing

Unit testing is performed to verify the functionality of individual modules such as the network scanning module, vulnerability detection module, and attack path generation module. Each module is tested independently to ensure that it performs its intended task correctly. This helps in identifying errors at an early stage and improves code reliability.

9.2.2 Integration Testing

Integration testing ensures that different modules of the system work together seamlessly. It verifies the interaction between components such as scanning, vulnerability detection, and attack path generation. This ensures smooth data flow across the system and confirms that all modules are properly connected.

9.2.3 System Testing

System testing evaluates the complete framework as a whole. It checks whether all components work together efficiently under real-time conditions. This testing ensures that the system meets all functional requirements and performs as expected when provided with real inputs.

9.2.4 Security Testing

Security testing is performed to evaluate the system's ability to detect vulnerabilities and potential threats. Various scenarios such as open ports, weak services, and misconfigurations are analyzed. This ensures that the system is robust and capable of identifying security risks effectively.

9.3 TEST CASES

9.3.1 Network Scan Test

The system was tested using valid IP addresses and domains as input. The scanning module successfully identified open ports and running services. The results were displayed correctly, confirming that the scanning process works efficiently.

9.3.2 Vulnerability Detection Test

The system analyzed detected services and identified potential vulnerabilities using multiple techniques. It successfully detected insecure protocols and misconfigurations. This confirms that the vulnerability detection module functions accurately.

9.3.3 Attack Path Generation Test

The system generated attack paths based on detected vulnerabilities. It successfully mapped relationships between services and potential risks. The results were displayed clearly, demonstrating the effectiveness of the attack path generation module.

9.4 PERFORMANCE TESTING

Performance testing is conducted to evaluate the system's efficiency under different conditions. The framework is tested with multiple scan requests to check its ability to handle real-time operations. The system demonstrates stable performance and handles scanning processes efficiently.

The response time of the system is moderate due to real-time scanning operations. However, the attack path generation process is fast since it uses processed data. The system also shows good scalability, as it can handle increasing workloads without significant performance degradation.

9.5 RESULT ANALYSIS

The results obtained from testing show that the system performs effectively in detecting vulnerabilities and generating attack paths. The system was tested under various scenarios to evaluate its performance, reliability, and accuracy.

During the testing phase, the system successfully identified different types of security issues such as open ports, insecure protocols, and misconfigurations. The attack path generation module provided meaningful insights into potential attack scenarios.

The system demonstrates high reliability and efficiency. Although certain limitations exist, such as dependency on network conditions and incomplete service detection in some cases, the overall performance is satisfactory.

he framework is capable of providing valuable insights for cybersecurity analysis and can be effectively used in real-world environments.

CHAPTER 10

CONCLUSION AND FUTURE ENHANCEMENT

10.1 Conclusion

The Attack Path Discovery Framework is an advanced and intelligent system designed to automate vulnerability assessment and provide a clear understanding of how an attacker can exploit system weaknesses through multi-step attack paths. The system integrates multiple cybersecurity tools such as Nmap, OpenVAS, Searchsploit, LinPEAS, and BloodHound to collect, analyze, and correlate security data in a structured manner.

By accepting a target IP address or domain as input, the system performs network scanning to identify open ports and services, maps vulnerabilities using CVE databases, and identifies available exploits without executing them. It further analyzes possible privilege escalation opportunities and correlates all findings to generate realistic attack paths. These attack paths help in understanding how individual vulnerabilities can be chained together to compromise a system.

A key strength of the system lies in its ability to visualize attack paths using a graph-based approach with Neo4j. This allows security analysts and users to clearly see relationships between system components, vulnerabilities, and attack steps. The interactive visualization improves decision-making and helps prioritize security measures effectively.

Another important aspect of the system is its emphasis on ethical and safe practices. The framework does not perform real exploitation but only simulates attack scenarios, making it suitable for educational environments, research purposes, and controlled security assessments.

Overall, the Attack Path Discovery Framework provides a comprehensive solution for modern cybersecurity challenges by combining automation, correlation, and visualization. It enhances the understanding of system vulnerabilities, reduces manual effort, and supports proactive security analysis. The system demonstrates how integrating multiple tools and technologies can create a powerful platform for identifying and mitigating potential security risks.

10.2 Future Enhancement

Although the system provides an effective solution for attack path discovery, several enhancements can be implemented to improve its performance, accuracy, and usability.

1. Real-Time Threat Intelligence Integration

The system can be enhanced by integrating real-time threat intelligence feeds. This will allow the framework to:

- Detect newly discovered vulnerabilities
- Update exploit databases automatically
- Improve accuracy of attack path generation

2. AI-Based Risk Prediction

Artificial Intelligence and Machine Learning models can be incorporated to:

- Predict the most probable attack paths
- Prioritize vulnerabilities based on risk
- Provide intelligent recommendations

3. Cloud and Enterprise Integration

The system can be extended to support:

- Cloud environments (AWS, Azure, GCP)
- Large-scale enterprise networks
- Distributed systems

4. Automated Reporting System

Future versions can include:

- PDF report generation
- Detailed security assessment reports
- Customizable reporting templates

5. Enhanced User Interface

The frontend can be improved by:

- Adding real-time scanning progress indicators
- Providing interactive dashboards
- Improving graph visualization features

6. Integration with Security Tools

The system can be integrated with:

- SIEM tools (Splunk, ELK)
- Intrusion Detection Systems (IDS)
- Security Operation Center (SOC) platforms

7. Advanced Attack Simulation

Future improvements may include:

- Simulated attack execution in sandbox environments
- More detailed attack chain modeling
- Scenario-based risk analysis

8. Mobile and Web Deployment

The system can be deployed as:

- A cloud-based web application
- A mobile application for monitoring

9. Support for Additional Protocols and Systems

Enhancements can include:

- IoT security analysis
- Industrial control systems (ICS)
- Wireless network attack path analysis

CHAPTER 11

REFERENCES

Here are some relevant references that support the concepts used in this project:

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CHAPTER 12

CHART

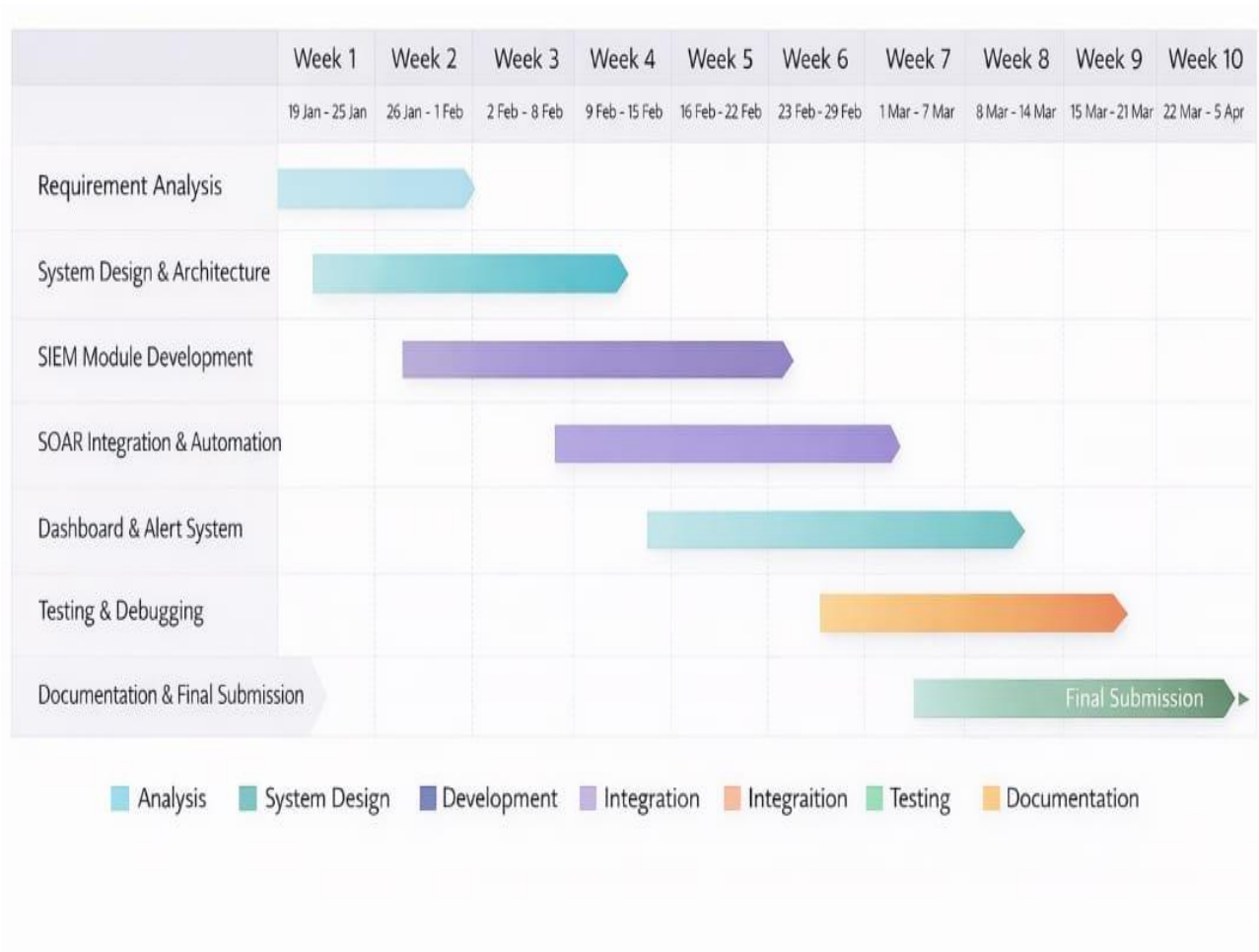


Fig 12.1 Gantt Chart



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